

Sulfate Radical-Mediated Degradation of Sulfadiazine by CuFeO₂ Rhombohedral Crystal-Catalyzed Peroxymonosulfate: Synergistic Effects and Mechanisms

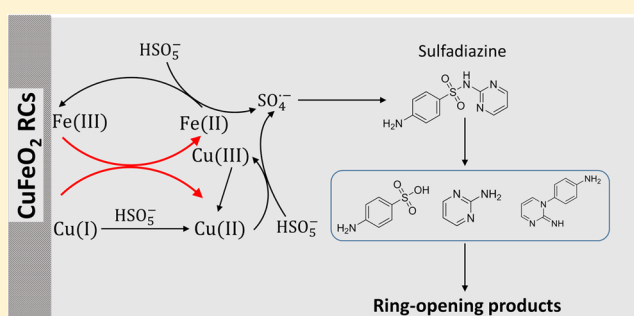
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S Supporting Information

ABSTRACT: Copper–iron bimetallic oxides have shown great potential for powerful radical production by activating peroxides. In this work, CuFeO₂ rhombohedral crystals (RCs) were synthesized and used as heterogeneous catalysts for peroxymonosulfate (PMS) activation under various conditions. Sulfadiazine, a widely used veterinary sulfonamide, was used as a target pollutant to evaluate the efficiency of this combination. The results showed that of all the catalysts tested, the CuFeO₂ RCs had the greatest reactivity. Under conditions of 0.1 g L⁻¹ CuFeO₂ RCs and 33.0 μM PMS, the nearly complete degradation of sulfadiazine occurred within 24 min. A synergistic catalytic effect was found between solid Cu(I) and Fe(III), probably due to the accelerated reduction of Fe(III). The two activation stages that produced different radicals (hydroxyl radicals followed by sulfate radicals) existed when solid Cu(I) was used as the catalyst. The CuFeO₂ RCs had a higher PMS utilization efficiency than CuFe₂O₄, probably because the Cu(I)-promoted reduction of solid Fe(III). A total of 10 products were identified, and their evolution was explored. On the basis of the evidence of oxidative product formation, we proposed four possible pathways of sulfadiazine degradation.



INTRODUCTION

Sulfate radical (SO₄^{•-})-based advanced oxidation processes have received a great deal of attention from environmental scientists for their use in wastewater purification.^{1,2} SO₄^{•-}, with a redox potential of 2.5 to 3.1 V,³ has the potential of becoming an alternative to the hydroxyl radical (•OH), probably due to its high selective oxidation, independence of pH, and strong oxidizing capability.^{1,4} Generally, peroxymonosulfate (PMS; HSO₅⁻) and persulfate are the two peroxides used for the production of SO₄^{•-}. Compared with persulfate, PMS is generally more easily activated by transition metals.⁵ In fact, various metals, such as Co²⁺, Mn²⁺, Fe²⁺, and Ce³⁺, have been tested to activate PMS over the past few years.^{6–8} Of all the metals evaluated, Co²⁺ has the highest reactivity for PMS activation.² However, due to its potential carcinogenic effects, using cobalt as a catalyst has limitations in practical applications. Consequently, the development of efficient catalysts for PMS activation remains a priority.

As a solid catalyst, CuO has shown reactivity in activating PMS oxidation of phenol.⁹ However, CuO is not stable, especially under acidic conditions, and this limits its potential for water treatment applications. Recently, copper–iron bimetallic catalysts have shown great promise in the catalysis of peroxides. Magnetic CuFe₂O₄ and its derivative composites

were synthesized and used for the production of SO₄^{•-} and •OH by activating PMS,^{10,11} H₂O₂,^{12–14} or persulfate.¹⁵ In these experiments, pollutants were efficiently degraded and even mineralized. In addition, the bimetallic oxides revealed greater reactivity than their corresponding single-metal-containing oxides¹⁰ and displayed higher stability than CuO, probably due to the formation of Cu–Fe bonds in the bimetallic structure.

Compared with Cu²⁺, Cu⁺ has received significantly less attention, especially its solid form, ≡ Cu(I) (≡ is used to represent the element in solid form). Zhang et al.¹⁶ used CuFeO₂ to degrade bisphenol A with H₂O₂ as the oxidant and found that the degradation of bisphenol A with CuFeO₂ was over five times greater than that with nano Fe₃O₄ particles. In addition, the level of copper leached from CuFeO₂ was just 10% of that leached from Cu₂O. These findings suggest that as a catalyst, CuFeO₂ has great efficiency and high stability. However, its reactivity toward PMS is still unknown. In contrast, compared with iron, copper is a relatively new catalyst,

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and its catalytic mechanism is still not quite clear. Some studies on the activation of H_2O_2 by Cu^{2+} have shown that Cu^{2+} probably first undergoes a reduction step that generates Cu^+ , which then reacts with H_2O_2 to produce oxidizing species.^{12,17–19} However, during interaction with PMS, oxidation rather than reduction of Cu^{2+} probably takes place in the first step.^{10,15} Because both H_2O_2 and PMS can act as either oxidants or reductants and have a similar reduction potential (1.77 V for H_2O_2 and H_2O and 1.80 V for HSO_5^- and HSO_4^-), complicated conversion cycles of copper might exist during the activation processes that deserve further investigation.

In this study, cuprous-iron bimetallic oxide CuFeO_2 rhombohedral crystals (RCs) were synthesized with a hydrothermal method and used as catalysts for PMS activation, with sulfadiazine (SDZ; 4-amino-N-(2-pyrimidinyl) benzenesulfonamide), a widely used veterinary sulfonamide, as the target contaminant. The reactivity of CuFeO_2 RCs was compared with that of other catalysts and investigated under various conditions. A synergistic effect between $\equiv \text{Cu(I)}$ and $\equiv \text{Fe(III)}$ was identified for the first time, which suggests an innovative solution for promoting the circulation of $\equiv \text{Fe(III)}$ and $\equiv \text{Fe(II)}$. Possible pathways for the degradation of SDZ were proposed, and the mechanism for PMS activation on the surface of the CuFeO_2 RCs was also investigated.

■ EXPERIMENTAL SECTION

Chemicals and Reagents. Unless otherwise specified, all chemicals and reagents were of reagent grade and used as received from Sigma-Aldrich. Oxone ($2\text{KHSO}_5 \cdot \text{KHSO}_4 \cdot \text{K}_2\text{SO}_4$, 4.7% active oxygen) was purchased from J & K Scientific (Hong Kong), liquid chromatography (LC)/mass spectrometry (MS)-grade formic acid was purchased from Waters Corp. (Milford, MA), and Optima LC/MS-grade methanol was provided by Fisher Scientific (Pittsburgh, PA). All solutions were prepared with purified water drawn from a Barnstead EASYpure UV/UF water purification system.

Synthesis of CuFeO_2 RCs and CuFe_2O_4 . CuFeO_2 RCs were synthesized using a modified hydrothermal method according to a previously reported protocol.²⁰ Copper(II) nitrate and iron(III) nitrate were used as sources of copper and iron. In a typical procedure, 0.02 M $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ and 0.02 M $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$ were dissolved in 80 mL of pure water, and 0.2 M NaOH was added to the solution and stirred for 10 min to form a suspension. Then, 1 mL of propionaldehyde, the reducing agent, was introduced into the suspension and stirred for another 5 min. The resulting mixture was transferred to a 100 mL Teflon-lined stainless steel autoclave and put in an oven at 180 °C for 24 h. The ratio of copper and iron in the catalysts was determined to be 1:1 after the catalysts were completely dissolved using nitric acid and hydrofluoric acid. Magnetic CuFe_2O_4 particles were synthesized with a modified sol-gel combustion method using citrate as the complex agent.²¹ Details are listed in Text S1.

Experimental Procedure. Unless otherwise specified, all of the degradation experiments were carried out in 50 mL of polypropylene copolymer Nalgene centrifuge tubes (Rochester, NY) containing 20 mL of 8.0 μM SDZ solution (with 10 mM phosphate buffer at pH 6.8) at room temperature (25 ± 2 °C). The pH variation was within ± 0.05 throughout the experiments. When investigating the effects of solution pH, the initial pH values of the SDZ solution were adjusted with 0.1 M NaOH or 0.1 M HClO_4 . Polypropylene copolymer tubes were selected for their low adsorption toward pollutants, as

demonstrated in our previous study.²² The catalysts were first added to the tubes, followed by PMS from a stock solution to initiate the degradation. During the investigation of the effects of bicarbonate and chloride, certain amounts of NaHCO_3 and NaCl were first added to the SDZ solution, followed by pH adjustment and addition of catalysts and PMS. The tube reactors were then placed on an orbital shaker at 120 rotations per min. Samples were withdrawn with syringes at each sampling point and filtered with 0.22 μm polytetrafluoroethylene syringe filters (Millipore, German). The filtration had no obvious impact on the SDZ concentration. The filtered samples (1 mL) were transferred to 2 mL autosampler vials, and 100 μL of ethanol, a quenching agent, was immediately added to the vials. To prevent the SDZ from further degradation, we analyzed the resulting samples within 20 min. The reactors were sacrificial with at least two reactors sampled for each time point, and the data obtained were averaged together. To test the reusability and stability of the CuFeO_2 RCs, we collected the particles by centrifugation at 4000 rotations per min for 3 min, washed several times with purified water, and dried in a vacuum-drying oven at ambient temperature. The results are presented in Figures S12 and S13.

Analytical Methods. The SDZ concentration was measured with a Waters Acquity ultraperformance LC (UPLC) system equipped with a photodiode array detector at a wavelength of 268 nm. Details are shown in Text S2. The pH values were determined with an Orion 2-Star benchtop pH meter (Thermo Scientific). The metal ions were quantified by an Agilent 7700 inductively coupled plasma-mass spectrometer. To analyze the degradation products of SDZ, we used UPLC coupled with tandem mass spectrometry (MS/MS, Waters Acquity TQD). The MS was operated in full-scan MS, selected-ion recording, and multiple-reaction-monitoring modes with electrospray ionization as the ion source. Details are described in Text S3.

Characterization of Catalysts. The X-ray powder diffraction spectra of the catalysts were recorded on a Bruker D8 Advance X-ray powder diffractometer (Bruker Co., Ltd.) with a $\text{Cu K}\alpha_{1,2}$ X-ray radiation source operated at 40 kV and 40 mA (Text S4 and Figure S1a–f). The surface morphology of the catalysts was characterized with a Hitachi S-4800 field emission scanning electron microscope and an FEI Tecnai G2 20 S-TWIN high-resolution transmission electron microscope (Text S4 and Figure S1a–c). The valence variation of copper was examined with an ESCALAB 250XI X-ray photoelectron spectroscopy (Thermo) with $\text{Al K}\alpha$ radiation. The binding energy was calibrated using the C 1s peak at 284.8 eV. The Brunauer–Emmett–Teller specific surface area and average pore size were determined through N_2 adsorption and desorption experiments conducted using a Beckman Coulter SA 3100 surface analyzer. The average particle size was measured with a Mastersizer 3000 laser particle size analyzer. The pH_{pzc} (point of zero charge) was determined by acid–base titration experiments. All of the results are listed in Table S1. The attenuated total reflectance Fourier transform infrared spectroscopy (ATR-FTIR) spectra of the catalysts were recorded according to the procedures described by Zhang et al.¹⁰ Details are shown in Text S5.

■ RESULTS AND DISCUSSION

Reactivity of the CuFeO_2 RCs. The CuFeO_2 RCs had no obvious adsorption toward SDZ (Figure S3). The catalytic performance of the CuFeO_2 RCs was compared with that of

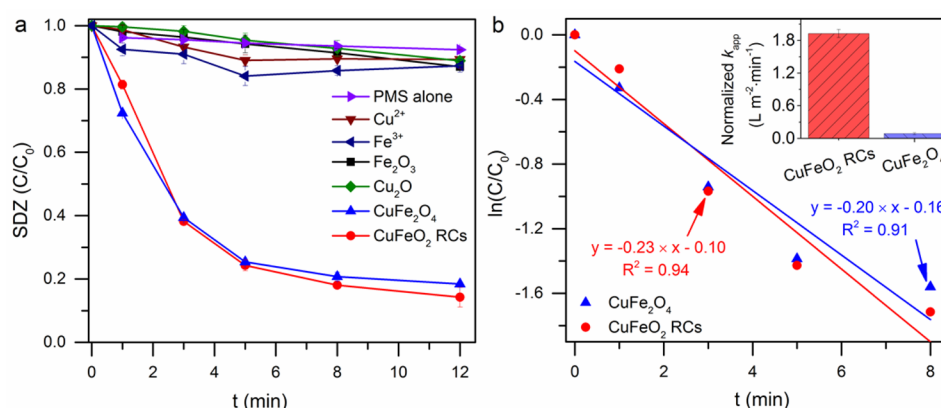
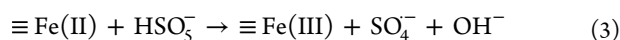
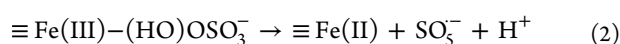
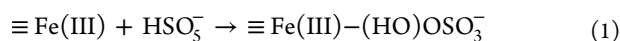


Figure 1. SDZ degradation by PMS with different catalysts (a) and plot of $\ln(C/C_0)$ versus reaction time (apparent degradation rate constant k_{app} (b)). The inset in (b) shows the specific-surface-area normalized k_{app} values. Symbols represent the mean value from triplicate experiments, and solid lines represent the linear regression. Conditions: [SDZ] = 8.0 μM , [PMS] = 33.0 μM , $[\text{Cu}^{2+}] = [\text{Fe}^{3+}] = 0.05 \text{ g L}^{-1}$, [solid catalysts] = 0.1 g L^{-1} , and 10 mM phosphate buffer at pH 6.8.

other catalysts (Figure 1a). In the presence of 10 mM phosphate buffer, only a slight degradation of SDZ was noticed when 50 mg L^{-1} Cu^{2+} or 50 mg L^{-1} Fe^{3+} was present. This observation suggested that the degradation of SDZ due to the ion-leaching-induced homogeneous catalytic reactions was negligible when solid copper or iron-containing oxides were used as catalysts. In addition, no obvious promotion effect was noticed when Fe_2O_3 was added to SDZ solutions in the presence of PMS, indicating that $\equiv\text{Fe(III)}$ had low reactivity for PMS activation. This finding was consistent with previous reports, in which the reduction of $\equiv\text{Fe(III)}$ was found to be the rate-determining step in $\text{SO}_4^{\bullet-}$ generation.²³ In the process of interacting with PMS, $\equiv\text{Fe(III)}$ was first reduced to $\equiv\text{Fe(II)}$ (eqs 1 and 2),¹¹ which then reacted with PMS to produce $\text{SO}_4^{\bullet-}$ (eq 3). The slow reduction of $\equiv\text{Fe(III)}$ was probably due to the fact that HSO_5^- (1.8 V) had a much higher reduction potential than Fe^{3+} (0.77 V), which made the redox thermodynamically unfavorable.



Similar to Fe_2O_3 , Cu_2O also had low efficiency for PMS activation in the presence of phosphate buffer, which had little influence on its catalytic reactivity (Figure S4a). Because the reaction between Cu_2O and PMS produces $\cdot\text{OH}$,^{10,24} and $\cdot\text{OH}$ reacts with Cu^+ at a constant rate of $\sim 1.0 \times 10^{10} \text{ M}^{-1} \text{ s}^{-1}$,¹⁶ the low efficiency of Cu_2O was probably due to the scavenger effect of $\equiv\text{Cu(I)}$. In contrast, a significant improvement in the degradation was observed when bimetallic catalysts were used. Compared to CuFe_2O_4 , the CuFeO_2 RCs had even greater catalytic reactivity. Under the conditions of 0.1 g L^{-1} CuFeO_2 RCs and 33.0 μM PMS, approximately 86% of the SDZ could be degraded. In the absence of phosphate buffer, the reactivity of the CuFeO_2 RCs was increased, realizing the nearly complete degradation of SDZ after 12 min (Figure S4b).

The heterogeneous catalytic decomposition of peroxides is controlled by intrinsic reactions on the catalyst surface.^{25,26} These intrinsic reactions are mediated by available reactive sites that are directly proportional to the specific surface area of the catalysts. Therefore, to better evaluate the catalytic reactivity, pseudo-first-order kinetics were used to model the degradation

of SDZ. The pseudo constants obtained were normalized by specific surface area per volume under catalyst loading of 0.1 g L^{-1} . Before normalization, the pseudo constant for the CuFeO_2 RCs (0.23 min^{-1} , Figure 1b) was slightly higher than that for CuFe_2O_4 (0.20 min^{-1} , Figure 1b). After normalization, however, the constant for the CuFeO_2 RCs (2.0 $\text{L m}^{-2} \text{ min}^{-1}$) was more than 20 times higher than that for CuFe_2O_4 (0.09 $\text{L m}^{-2} \text{ min}^{-1}$), indicating the active sites on the CuFeO_2 surface are more reactive. Because Cu_2O –PMS and Fe_2O_3 –PMS were relatively low in efficiency for SDZ degradation (Figure 1a), the high reactivity of the CuFeO_2 RCs was probably due to the potential synergistic effect between $\equiv\text{Cu(I)}$ and $\equiv\text{Fe(III)}$.

Behavior of $\equiv\text{Cu(I)}$ and Synergistic Effect. To explore the catalytic mechanism of $\equiv\text{Cu(I)}$, we further examined the catalytic behavior of Cu_2O and Fe_2O_3 (Figure 2a). During the initial reaction stage (first 20 min), no obvious difference was observed between the degradation of SDZ by Cu_2O –PMS and that by Fe_2O_3 –PMS. As the reaction continued, however, the degradation by Cu_2O –PMS was accelerated and displayed greater overall efficiency than that by Fe_2O_3 –PMS. During the first 20 min, approximately 22% of the SDZ was degraded by Cu_2O –PMS, whereas an overall efficiency of around 57% was realized after 40 min, suggesting that 35% of the SDZ was degraded during the final 20 min. The kinetic investigations showed that the pseudo-first-order degradation constants for SDZ degradation in Cu_2O –PMS were calculated to be 0.012 and 0.029 $\text{M}^{-1} \text{ s}^{-1}$ for the first and final 20 min, respectively (Table S2). In the process of reaction with PMS, $\equiv\text{Cu(I)}$ was oxidized to $\equiv\text{Cu(II)}$, accompanied by the generation of $\cdot\text{OH}$ (eq 4).^{10,24} Previous investigations have shown that $\equiv\text{Cu(II)}$ becomes active for PMS activation by first forming a surface complex $\equiv\text{Cu(II)}-(\text{HO})\text{OSO}_3^-$ with HSO_5^- via inner-sphere complexation (eq 5).¹⁰ This complex then decomposes into $\equiv\text{Cu(III)}$ and $\text{SO}_4^{\bullet-}$ (eq 6). The standard reduction potential of $\equiv\text{Cu(III)}-\equiv\text{Cu(II)}$ in solid form has been reported to be 2.3 V,²⁷ much higher than the potential of $\equiv\text{Fe(III)}-\equiv\text{Fe(II)}$ (0.77 V) or HSO_5^- – HSO_4^- (1.80 V). Thus, the $\equiv\text{Cu(III)}$ produced was likely reduced by both HSO_5^- (eqs 7 and 8) and $\equiv\text{Fe(II)}$ (eq 9) in the oxidation systems of CuFeO_2 RCs–PMS and CuFe_2O_4 –PMS. The experiments in which CuO was used as the catalyst confirmed that this oxide had greater reactivity than Cu_2O (Figure S4c), suggesting that the

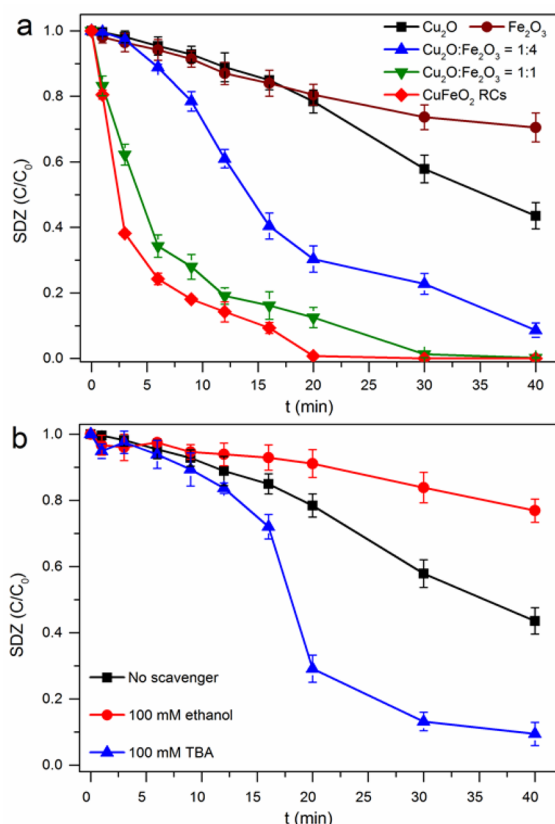
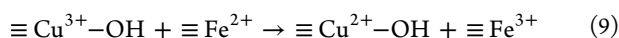
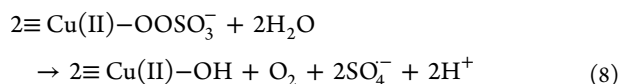
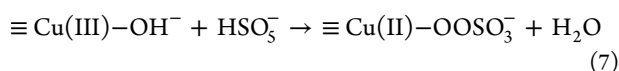
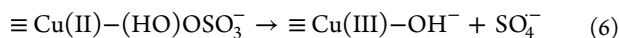
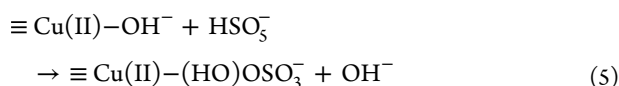
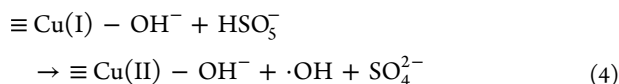


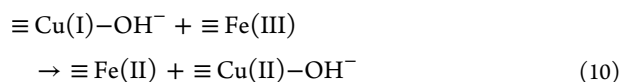
Figure 2. SDZ degradation by PMS with Cu₂O–Fe₂O₃ mixtures as the catalysts (a). Effect of radical scavengers on the degradation performance of Cu₂O–PMS (b). Conditions: [catalysts] = 0.1 g L^{−1}, [PMS] = 33.0 μM, [SDZ] = 8.0 μM, and 10 mM phosphate buffer at pH 6.8.

fast degradation observed during Cu₂O–PMS oxidation in the later reaction stage was probably due to the formation of ≡ Cu(II) on the surface of Cu₂O, which was further verified by the XPS spectra for Cu 2p of the fresh and used Cu₂O catalysts (Figure S5).



To explore the potential synergistic effect, we used two mixtures (with weight ratios between Cu₂O and Fe₂O₃ of 1:4 or 1:1) as catalysts. Theoretically, the reactivity of these mixtures should lie between that of Cu₂O and Fe₂O₃, suggesting that the degradation of SDZ in the presence of both Cu₂O and Fe₂O₃ should never exceed 57% after 40 min. However, as shown in

Figure 2a, these two mixtures were much more reactive than both Cu₂O and Fe₂O₃, and an increase of Cu₂O in the mixture further enhanced the degradation (Figure 2a). With 0.1 g L^{−1} Cu₂O and Fe₂O₃ (1:1), approximately 98% of the degradation could be achieved after 30 min. Therefore, there was a synergistic effect between ≡ Cu(I) and ≡ Fe(III) during the activation of PMS. Given the facts that the reduction of ≡ Fe(III) is the rate-limiting step of SO₄^{•−} generation, and Cu₂O–PMS is inefficient for SDZ degradation (Figure 1a), this synergistic effect probably resulted from the redox reaction between ≡ Cu(I) and ≡ Fe(III) (eq 10), which was further verified by the absence of synergistic effect between CuO and Fe₂O₃ (Figure S6). However, the molar ratio between ≡ Cu(I) and ≡ Fe(III) in the Cu₂O–Fe₂O₃ mixture (1:1) was 1:1, which was the same as that in the CuFeO₂ RCs. However, the reactivity of this mixture was still obviously lower than that of the CuFeO₂ RCs (Figure 2a), probably due to the formation of Cu–Fe bonds that were beneficial for electron transfer between ≡ Cu(I) and ≡ Fe(III). In addition, ~220 μg L^{−1} copper ions were detected in the Cu₂O–Fe₂O₃ mixture (1:1) suspension after 40 min, whereas only ~21 μg L^{−1} of copper ions were determined in the case of CuFeO₂ RCs (Figure S7). These leaching results suggest that the CuFeO₂ RCs had much higher stability than the Cu₂O–Fe₂O₃ mixture, and the potential second pollution with the CuFeO₂ RCs due to metal leaching is not a concern.



Identification of Radicals. ·OH can be generated by either the reaction between ≡ Cu(I) and PMS^{10,26} or the reaction between SO₄^{•−} and H₂O.^{11,28} Thus, both ·OH and SO₄^{•−} were probably present in CuFeO₂ RCs–PMS oxidation. It is widely accepted in the literature that alcohols with an α hydrogen such as ethanol readily react with both ·OH and SO₄^{•−} radicals.^{6,28} Alcohols without an alpha hydrogen, such as *tert*-butyl alcohol (TBA), are also readily reactive toward ·OH, but their reaction with SO₄^{•−} is over 1000-fold slower (Table S3). Thus, ethanol and TBA were usually used to differentiate SO₄^{•−} from ·OH. The effects of ethanol and TBA on CuFeO₂ RCs–PMS oxidation (Table 1) showed that ethanol had a much higher

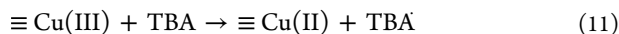
Table 1. Inhibition Percentage in SDZ Degradation When Specific Alcohols Are Present in Solution^a

alcohol concentration (mM)	ethanol	TBA
10	(41.3 ± 1.9)	(6.0 ± 1.1)
100	(67.7 ± 5.5)	(8.7 ± 0.7)
500	(82.4 ± 2.2)	(19.8 ± 1.9)

^aConditions: [SDZ] = 8.0 μM, [PMS] = 33.0 μM, [CuFeO₂] = 0.1 g L^{−1}, and 10 mM phosphate buffer at pH 6.8.

inhibition effect on the degradation of SDZ compared with TBA. For instance, SDZ degradation decreased by ~68% in the presence of 100 mM ethanol in comparison with the control experiment, in which no scavenger was present, whereas only around 9% inhibition was observed in the presence of the same amount of TBA. Therefore, SO₄^{•−} was mainly responsible for the degradation of SDZ during CuFeO₂ RCs/PMS oxidation. However, the effect of ethanol on Cu₂O–PMS showed that the degradation of SDZ was also greatly inhibited, as the final degradation rate decreased from 57% to 23% (Figure 2b). In

contrast, when TBA was used as the scavenger, a significant promotion effect was observed. With the addition of TBA, the final degradation increased to ~90%. Because TBA was highly reactive toward $\cdot\text{OH}$ and inert for $\text{SO}_4^{\cdot-}$, $\text{SO}_4^{\cdot-}$ was probably also mainly responsible for the degradation of SDZ during Cu_2O –PMS oxidation. This conclusion is consistent with the low efficiency of Cu_2O on PMS observed during the initial reaction stage (Figure 2a) and the quenching results obtained with CuFeO_2 RCs (Table 1) as the catalysts. According to recent findings from W.T. David et al.,¹⁷ TBA is a strong reducing agent for $\equiv\text{Cu(III)}$, with a second-order rate constant of $\sim 1 \times 10^7 \text{ M}^{-1} \text{ s}^{-1}$. Thus, the promotion effect of TBA observed in Figure 2b was probably related to the accelerated reduction of $\equiv\text{Cu(III)}$ to $\equiv\text{Cu(II)}$ by TBA (eq 11), which was further supported by the promotion and inhibition effects of TBA on CuO –PMS and Fe_2O_3 –PMS systems, respectively (Figure S8). The effects of TBA concentration showed that the degradation of SDZ in Cu_2O –PMS in the final stage was greatly accelerated with the increase of TBA for 0 mM to 100 mM (Table S2), which is consistent with the assumption that the generated Cu(II) on the surface of Cu_2O was probably responsible for the faster degradation of SDZ.



Effects of Catalyst Loading and Temperature. Experiments were carried out to explore the effects of catalyst loading on SDZ degradation in the oxidations of CuFeO_2 RCs–PMS (Figure 3a) and CuFe_2O_4 –PMS (Figure 3b). An increase in CuFeO_2 RC loading within the tested range had an obvious positive effect on SDZ degradation, whereas a similar relationship was observed only when CuFe_2O_4 loading was between 0.01 and 0.1 g L^{-1} . When its loading increased from 0.1 to 0.3 g L^{-1} , no obvious improvement in degradation

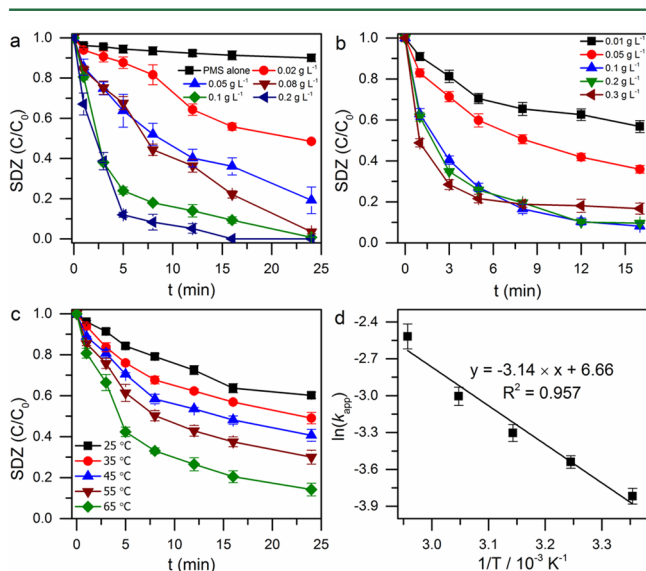


Figure 3. Effect of catalyst loading on SDZ degradation in CuFeO_2 RCs–PMS (a) and CuFe_2O_4 –PMS (b). The effect of reaction temperature (c) on SDZ degradation during CuFeO_2 RCs–PMS oxidation. (d) Plot of $\ln(k_{\text{app}})$ versus $1/(T \times 10^{-3})$ in CuFeO_2 RCs–PMS oxidation. Symbols represent the mean value from triplicate experiments, and solid lines represent the linear regression. Conditions: $[\text{PMS}] = 33.0 \mu\text{M}$, $[\text{CuFeO}_2 \text{ RCs}] = 0.01 \text{ g L}^{-1}$ (c), $[\text{SDZ}] = 8.0 \mu\text{M}$, and 10 mM phosphate buffer at pH 6.8.

performance occurred; in contrast, a slight decline in the final degradation after 16 min was noticed, suggesting that the PMS dose was the controlling factor of CuFeO_2 RCs/PMS oxidation. Considering the reactive properties of $\equiv\text{Fe(II)}$ toward the radicals ($\cdot\text{OH}$ and $\text{SO}_4^{\cdot-}$) produced in the oxidation, this decline was probably due to the scavenger effect of the catalysts. Similar phenomena were also reported by Rastogi et al.²⁹ Considering the observations in Figure 3a,b, the CuFeO_2 RCs had higher stoichiometric efficiency of PMS (amount of SDZ degraded per unit of PMS) than that of CuFe_2O_4 . The high efficiency of PMS in the presence of the CuFeO_2 RCs was probably due to the synergistic effect between $\equiv\text{Cu(I)}$ and $\equiv\text{Fe(III)}$, as a high dose of oxidants is usually needed to improve the performance of Fe(III) -promoted peroxide activation by accelerating the reduction of Fe(III) .³⁰ However, the nearly complete degradation of SDZ was realized after 24 min under conditions of 0.1 g L^{-1} CuFeO_2 RCs and 33.0 μM PMS, which suggests the strong degradation capability of CuFeO_2 RCs–PMS oxidation.

The degradation of SDZ increased with increasing reaction temperature (Figure 3c). The activation energy (E_a), which indicates the ease at which a reaction occurs, can be calculated with the Arrhenius eq (eq 12). According to the relationship between the pseudoconstants and reaction temperature (Figure 3d), the activation energy for SDZ degradation was 26.1 kJ mol^{-1} . This value is much higher compared with the activation energy of diffusion-controlled reactions, which typically ranges from 10 to 13 kJ mol^{-1} ,³¹ suggesting that the apparent reaction rate for SDZ degradation during CuFeO_2 RCs–PMS oxidation was dominated by the rate of intrinsic chemical reactions on the catalyst surface. In addition, because out-sphere interactions are usually diffusion-controlled reactions,³² the activation of PMS by the CuFeO_2 RCs was most likely an inner-sphere electron-transfer process. In contrast, the activation energy for SDZ degradation during CuFe_2O_4 –PMS oxidation was 36.5 kJ mol^{-1} (data not shown), which is much higher than the value obtained in the presence of CuFeO_2 RCs–PMS, indicating that the catalytic reactivity of CuFeO_2 RCs was much greater than that of CuFe_2O_4 .

$$\ln(k) = \ln(A) - \frac{E_a}{R} \left(\frac{1}{T} \right) \quad (12)$$

Effect of Solution pH, Ionic Strength, and Radical Scavengers. CuFeO_2 RCs–PMS oxidation showed good degradation of SDZ in the solution pH range of 6.8 to 10.0, with the best performance achieved at pH 8.5 (Figure 4a). However, its efficiency obviously decreased when the solution was adjusted to strongly acidic conditions (pH 4.0 and pH 5.0). According to the pK_a values of PMS (pK_{a1} is less than 0,³³ and pK_{a2} equals 9.4²⁸), PMS mainly exists as HSO_5^- when pH ranges from 4.0 to 8.5. The inhibition effect of H^+ could be explained by the stabilization effect of H^+ on the HSO_5^- .^{10,28} The high efficiency of CuFeO_2 RCs/PMS oxidation at neutral and alkaline conditions suggests that this process is suitable for the treatment of surface water.

Increasing ionic strength tends to influence the electrostatic interactions (outer-sphere interactions) in a heterogeneous system by compressing the thickness of the electric double layers. However, the inner sphere is not affected. The effect of ionic strength (represented as NaClO_4 concentration) (Figure 4b) shows that increasing NaClO_4 concentration from 0 to 100 mM had no obvious influence on the degradation of SDZ, which suggests that the electrostatic interactions between the

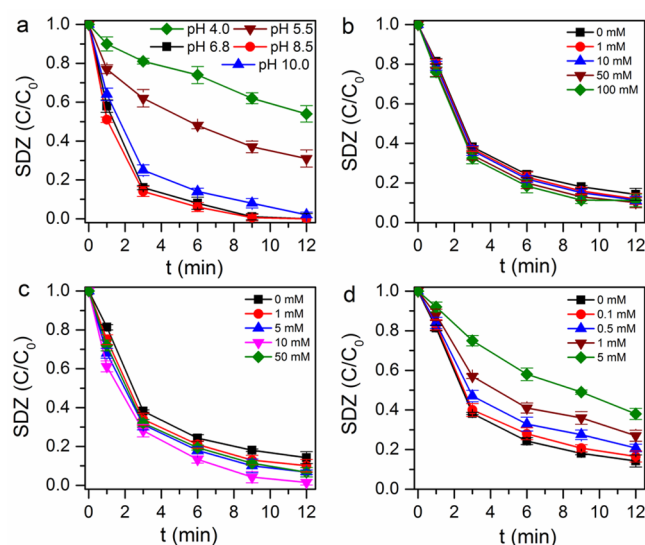


Figure 4. Effect of solution pH (a), ionic strength (represented as NaClO₄ concentration (b), bicarbonate (represented as NaHCO₃ concentration (c), and chloride (d) on SDZ degradation in CuFeO₂ RCs–PMS. Symbols represent the mean value from triplicate experiments. Conditions: [PMS] = 33.0 μM, [CuFeO₂ RCs] = 0.1 g L^{−1}, [SDZ] = 8.0 μM, and 10 mM phosphate buffer at pH 6.8 (b–d).

positively charged CuFeO₂ RCs and the negatively charged HSO₃[−] was not the dominant mechanism for radical production. Instead, the interactions probably proceed via the inner-sphere complexation.

Bicarbonate is widely present in surface water as the representative of inorganic carbon.³⁴ Generally, bicarbonate was reported to have scavenger effects in advanced oxidation processes.³⁵ However, an increase in the degradation of SDZ was observed when the concentration of bicarbonate in CuFeO₂ RCs–PMS was increased from 0 mM to 10 mM (Figure 4c). A further rise in bicarbonate concentration to 50 mM inhibited the degradation, but the SDZ degradation was still higher than the control experiment, in which no bicarbonate was present. Similar promotion effects were also noticed by Lei et al.¹⁵ when examining the influence of NaHCO₃ on the CuO–Fe₃O₄–persulfate system. The effect of bicarbonate on Cu(I) involves very complex processes, and the promotion effect of bicarbonate in Figure 4c is possibly due the formation of reactive Cu(I) carbonate complexes³⁶ and the potential activation reactivity of bicarbonate on peroxides.³⁷ However, detailed mechanisms under these processes are unclear and deserve further research. As a radical scavenger, chloride can consume SO₄^{•−}, accompanied by the generation of chlorine radicals.³⁸ Generally, chloride had inhibition effects on the degradation of SDZ in CuFeO₂ RCs and PMS when its concentration was increased from 0 to 5 mM (Figure 4d). Because chlorine radical (2.09 V) has a lower redox potential than SO₄^{•−} (2.5 to 3.1 V),³⁸ these inhibition effects were probably due to the consumption of SO₄^{•−} and the relative low reactivity of chlorine radical toward SDZ.

Mechanism for Surface Activation of PMS. Redox conversion of M⁽ⁿ⁺¹⁾⁺ (M represents a transition metal) to Mⁿ⁺ is involved in the activation of PMS by transition metals. It is difficult to find any direct experimental evidence regarding the formation of these species, probably due to the highly unstable properties of high-valent metals, such as Cu³⁺ and Fe⁴⁺. When metal oxides are mixed with water, the oxide surface tends to

strongly adsorb H₂O molecules. Due to the easy dissociation of the adsorbed H₂O into OH[−] and H⁺, hydroxyl groups prevalently exist on the surface sites as ligands.³⁹ High-valent M⁽ⁿ⁺¹⁾⁺ has a smaller ionic radius than low-valent Mⁿ⁺. Therefore, the binding force between M⁽ⁿ⁺¹⁾⁺ and surface ligands is larger than that between Mⁿ⁺ and surface ligands. On the basis of these theories, in situ spectroscopy has been used to indirectly identify the production of high-valent metals.^{10,15}

According to previous reports,^{10,40} IR at 1256 cm^{−1} can be assigned to the S–O stretching band of HSO₃[−] (Figure 5a). In the presence of CuFe₂O₄, this band blue-shifted 14 cm^{−1}, probably owing to the formation of the complex (≡ Cu(II)–(HO)OSO₃[−], eq 5) on the catalyst surface. Given the electron donation from ≡ Cu(II), OH was likely to have influenced the electron density of its neighboring bond, which caused the S–O stretching band to become stronger.^{10,15} In addition, the stretching band of the surface hydroxyl groups of CuFe₂O₄ occurred at ~2923 cm^{−1}. After catalytic reactions, this band became wide and shifted to ~3393 cm^{−1}. The peak intensity of this band also obviously increased. It has been proposed that surface hydroxyl groups are actively involved in the activation of PMS.⁴¹ The increase in these groups suggests the occurrence of strong hydroxylation during surface activation, which has been experimentally observed with X-ray photoelectron spectroscopies.^{13,16} Furthermore, ≡ Fe(III) in the spinel catalyst can act as a reservoir for the hydroxyl groups and donate them to a neighboring metal when reacting with PMS. Therefore, given that some part of ≡ Fe(III) was reduced to ≡ Fe(II), the blue-shift of these hydroxyl groups was probably due to the decrease in the electron withdrawing capability of the catalyst surface.

In contrast, a very slight red shift of the S–O band (from 1256 to 1251 cm^{−1}) was observed with the CuFeO₂ RCs (Figure 5b). As discussed above, the catalytic contribution from the ≡ Fe(III) in the CuFeO₂ RCs was greatly improved due to the presence of ≡ Cu(I). Thus, the ≡ Fe(III)–(HO)OSO₃[−] complex formed on the surface of the CuFeO₂ RCs actively participated in the reactions. Due to the strong electron attraction of ≡ Fe(III), the OH bonds on these metal sites decreased the electron density of neighboring S–O bonds, causing the HSO₃[−] stretching band to become weaker. Nevertheless, similar to the situation with CuFe₂O₄, the ≡ Cu(II)–(HO)OSO₃[−] complex also probably formed on the surface of CuFeO₂ RCs, and ≡ Cu(II) donated electrons to its neighboring OH. Therefore, the slight red shift was probably a combined result of these contradictory effects. The stretching band of the surface hydroxyl groups blue-shifted from 2994 to 3032 cm^{−1}. During CuFeO₂ RCs–PMS oxidation, part of ≡ Fe(III) and ≡ Cu(I) was converted to ≡ Fe(II) and ≡ Cu(II), respectively. These two conversions had contradictory effects on the binding force between the CuFeO₂ RCs and surface hydroxyl groups, which were probably the main reasons for the relatively small blue-shift.

Possible Pathways for SDZ Degradation. ·OH attacks organics via electron transfer, addition, or hydrogen abstraction,³² whereas SO₄^{•−} reacts more selectively, mainly through electron transfer.^{38,42} Therefore, different products are probably generated in the presence of these radicals. Although the degradation of sulfonamides by ·OH-dominated processes, such as photolysis,⁴³ photocatalysis,^{44–46} and Fenton reagents,^{47,48} has been widely reported, the transformation of sulfonamide by SO₄^{•−} has received much less attention. Thus, the degradation of SDZ was monitored with a UPLC–MS/MS system. As

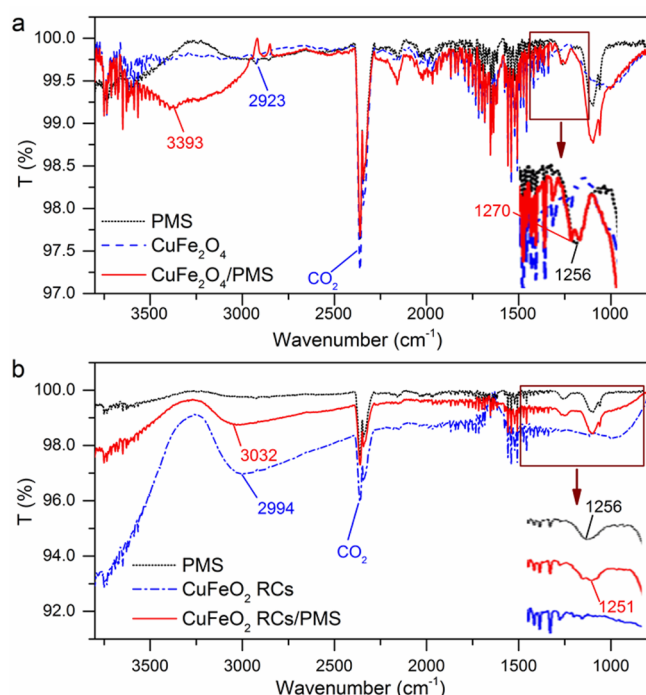


Figure 5. ATR-FTIR spectra of the PMS solution alone, bimetallic oxide particles in water, and bimetallic oxide particles in PMS solution at pH 6.8 without buffer. The H₂O spectrum has been subtracted from all above spectra. Conditions: [PMS] = 6.0 g L⁻¹ and [CuFe₂O₂ RCs] = [CuFe₂O₄] = 5.0 g L⁻¹.

shown in Table S4, 10 products were identified. On the basis of these products, we proposed four pathways for the degradation

of SDZ (Figure 6). The detection of intermediates with an m/z of 267 at 0.33 and 1.41 min showed that SDZ hydroxylation likely took place. Previous research has shown that sulfonamide hydroxylation can occur in the presence of either $\cdot\text{OH}$ or $\text{SO}_4^{\cdot-}$.^{48,49} Because $\text{SO}_4^{\cdot-}$ oxidizes compounds mainly via electron transfer, the hydroxylation of SDZ probably began with the generation of SDZ radicals ($\text{SDZ}^{\cdot+}$). These intermediates were highly unstable and quickly transformed into hydroxyl SDZ (product A, m/z 267) by reacting with H₂O via hydroxyl addition. Similar results have been reported by Gao et al.⁵⁰ and Ji et al.⁵¹ using $\text{SO}_4^{\cdot-}$ to degrade sulfamethazine and sulfamethoxazole, respectively.

SO_2 extrusion has been frequently reported during the degradation of sulfonamides.^{49,52} In the present study, an intermediate with an m/z of 187 (product B, $187 > 145$) was detected at 0.54 min (Table S4 and Figure S11a). The mass difference between SDZ and product B is 64, which is exactly equal to the SO_2 loss. Thus, compound B was assigned to the product of SO_2 extrusion. On the basis of the UPLC–MS/MS product ion (Figure S11b), we determined that this product was likely 4-[2-imino-pyrimidine-1(2H)-yl]aniline,⁵² which first accumulated and then decomposed as the reactions continued (Figure S10b). Cleavage of the S–N bond also likely occurred. During the degradation of SDZ, an intermediate with an m/z of 94 was found at 0.57 min (Table S4 and Figure S9a), and this was confirmed to be 2-aminopyrimidine (product C, $94 > 79$) with a concentration in the range of 4 to 8 $\mu\text{g L}^{-1}$ (Figure S10a). Generally, sulfanilic acid is thought to be produced from the determination of 2-aminopyrimidine. This acid is usually regarded as one of the products of S–N bond cleavage during sulfanilamide degradation, but some studies on sulfonamide mineralization have not successfully identified this com-

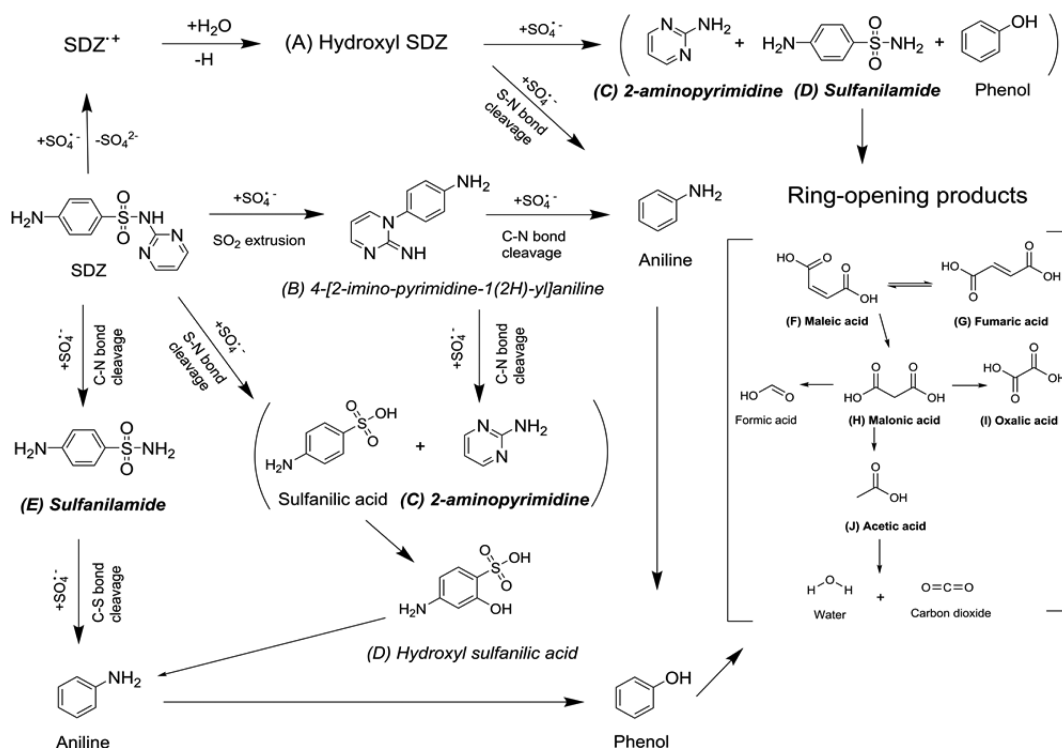


Figure 6. Proposed pathways for SDZ degradation during CuFe₂O₂ RCs–PMS oxidation. Compounds with capital letters were detected by UPLC–MS/MS. Bold names suggest products confirmed by reference standard compounds. Identified products with italic names are shown in Figures S9–S11 with further specifications.

pound.^{47,53} In this study, sulfanilic acid was also not found, but hydroxyl sulfanilic acid (product D, 191 > 123) appeared as a product at 0.39 min with a high signal-to-noise ratio of around 497 (Figure S11c) and a relatively constant concentration (Figure S10b). Therefore, sulfanilic acid was produced and quickly oxidized to hydroxyl sulfanilic acid. Sulfanilamide (product E, m/z 173 > 156) was detected at 0.52 min (Figure S9b). However, its signal was too low to be quantified. Based on the production of this product, it was proposed that the cleavage of the C–N bond linking pyrimidine and the benzene ring was also one of the pathways for SDZ degradation.^{47,54}

Under the attack of $\text{SO}_4^{\bullet-}$, phenol was likely generated as one of the common intermediates of products A, B, D, and E.^{49,50,55} This compound was readily degradable in advanced oxidation processes and was further decomposed into ring-opening products (Table S4),^{49,56} such as maleic acid (product F, 115 > 71) at 0.50 min and malonic acid (product H, 103 > 59) at 0.54 min. The successful detection of acetic acid (product J, 61 > 44) at 0.36 min indicated the strong mineralization capability of CuFe_2O_4 RCs–PMS oxidation.

■ ASSOCIATED CONTENT

■ Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.est.5b05974.

Text S1–S5: details regarding the procedures for CuFe_2O_4 preparation, SDZ analysis, degradation product identification, crystallinity and morphology, and ATR-FTIR characterization; Tables S1–S4: properties of the catalysts, rate constants for alcohol oxidation by radicals, effects of TBA, and products of SDZ degradation identified with UPLC-MS/MS; Figures S1–S13: characterization of the catalysts, SDZ adsorption, effect of phosphate buffer and TBA on SDZ degradation, XPS spectra of Cu 2p, UPLC-MS/MS multiple reaction monitoring chromatograms and product ion of some degradation products, concentration profile of SDZ and its degradation products, the reusability of the CuFe_2O_4 RCs, and XRD patterns of the used CuFe_2O_4 RCs. (PDF)

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Notes

The authors declare no competing financial interest.

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